

HDOC - Day 4 Assignment: Multiplication Without * Operator

Problem Statement:

Find the product of two integers without using the multiplication operator * by repeated addition, minimizing the number of additions, and correctly handling positive, negative, and zero conditions.

Task 1: Mistakes in the Given Algorithm

- 1. Negative Multiplier Failure:** When num2 is negative (e.g., 6×-5 or -9×-4), the initial loop condition $\text{index} \leq \text{num2}$ evaluates to false immediately because index starts at 1, causing the loop to never run and returning $\text{prod} = 0$.
- 2. Missing Zero-Check:** It lacks an immediate termination step when multiplying by zero, proceeding unnecessarily into index comparison logic.
- 3. Inefficient Additions:** The algorithm always loops num2 times. If $\text{num1} = 3$ and $\text{num2} = 12$, it performs 12 additions instead of 3 additions, violating the requirement of minimum possible additions.

Task 2: Corrected Algorithm

1. Start
2. Define num1, num2, a, b, prod, count, is negative as integers.
3. Input num1, num2.
4. If $\text{num1} == 0$ or $\text{num2} == 0$, set $\text{prod} = 0$ and jump to Step 11.
5. Determine Sign:

If $(\text{num1} < 0 \text{ AND } \text{num2} > 0) \text{ OR } (\text{num1} > 0 \text{ AND } \text{num2} < 0)$, set $\text{is negative} = 1$.

Else, set $\text{is negative} = 0$.
6. Take Absolute Values: $a = |\text{num1}|$, $b = |\text{num2}|$.

7. Optimize for Minimum Additions:

If $a < b$, swap a and b (ensures b is always $\leq a$, minimizing the loop counter).

8. Initialize: $prod = 0$, $count = 1$.

9. Loop for Repeated Addition:

While $count \leq b$:

$prod = prod + a$

$count = count + 1$

10. Apply Sign:

If is negative $== 1$, set $prod = -prod$.

11. Print $prod$.

12. Stop

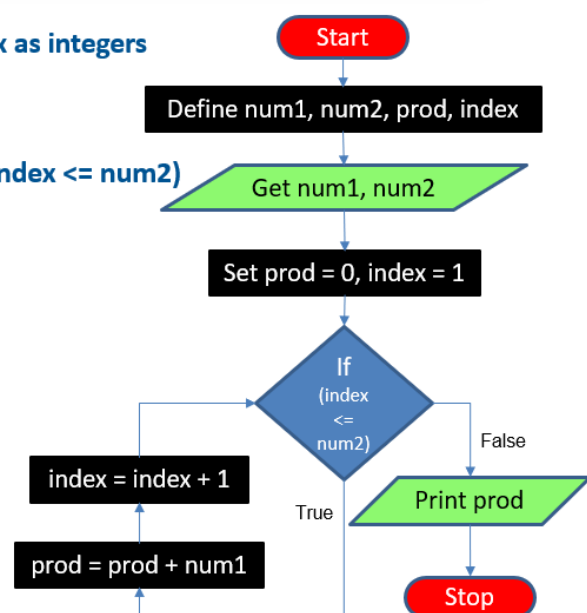
Task 3: Mistakes in the Given Flowchart



Multiply 2 numbers without using *operator (9)

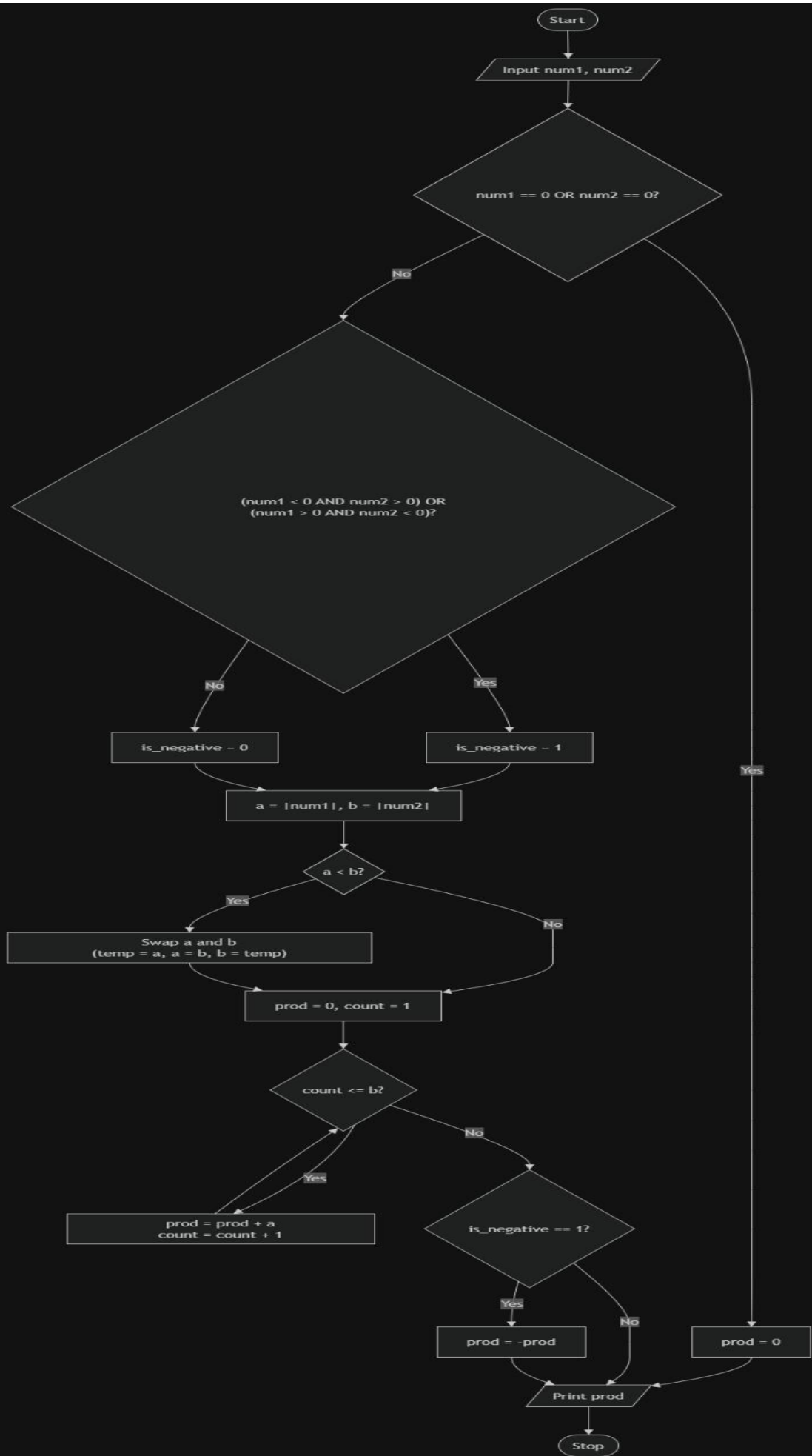
• Algorithm & Flowchart

1. Define $num1$, $num2$, $prod$, $index$ as integers
2. Get $num1$, $num2$
3. Set $prod = 0$, $index = 1$
4. Repeat steps 4.1 and 4.2 until ($index \leq num2$)
 - 4.1. $prod = prod + num1$
 - 4.2. $index = index + 1$
5. Print $prod$
6. Stop



1. Infinite / Skipped Loops on Negative Inputs: The decision diamond $\text{index} \leq \text{num2}$ fails when num2 is negative, causing false execution immediately without computing the product.
2. No Sign-Determination Logic: The flowchart has no branch to determine whether the output should be positive or negative based on operand signs.
3. Absence of Optimization Swap: It does not compare the magnitude of the two inputs to ensure that the smaller absolute value controls the loop counter.

Task 4: Corrected Flowchart:



Task 5: Verification Table with Minimum Additions

Test Case	Input	Expected Output	Minimum Additions	Actual Output	Status
1	12 3	Product: 36	3	Product: 36	Passed
2	3 12	Product: 36	3	Product: 36	Passed
3	-6 5	Product: -30	5	Product: -30	Passed
4	6 -5	Product: -30	5	Product: -30	Passed
5	-9 -4	Product: 36	4	Product: 36	Passed
6	-4 -9	Product: 36	4	Product: 36	Passed
7	0 8	Product: 0	0	Product: 0	Passed
8	8 0	Product: 0	0	Product: 0	Passed
9	0 -7	Product: 0	0	Product: 0	Passed
10	-7 0	Product: 0	0	Product: 0	Passed
11	0 0	Product: 0	0	Product: 0	Passed
12	1 -7	Product: -7	1	Product: -7	Passed
13	-1 -9	Product: 9	1	Product: 9	Passed